Simulating predictive-biomarker interactions in aggregated N-of-1 trials: a refined data-generating process following Hendrickson et al.

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Abstract

Background. The power of aggregated N-of-1 trials to validate a predictive biomarker is estimated by simulation. Hendrickson et al. (2020) introduced a framework for this, and were, to our knowledge, the first in the N-of-1 literature to encode the biomarker-by-treatment interaction as a correlation between the biomarker and the drug response within a joint multivariate normal distribution. We examine that encoding, correct its defects, and specify a refined data-generating process for trials of prazosin-like drugs.

Methods. We derive the moments and power of a paired-difference interaction statistic in closed form under the published process and refined versions of it. We verify them by Monte Carlo simulation, including runs of the published code. We compare covariance-based and mean-based binding of the biomarker, and stress-test analysis models against the features the refined process adds.

Results. As soon as any carryover is present, the published coupling rule gives every off-drug visit the full on-drug biomarker correlation. The interaction vanishes, and power falls to near the test size at every positive half-life. Coupling that decays with the stated carryover half-life restores a smooth decline, monotone in the half-life for the paired-difference statistic, and raised power in the published code from 0.13 to 0.64 at a half-life of 0.1 week. Conditional on the biomarker, covariance-based binding is mean moderation with a reduced residual variance, so mean-based binding that fades with the drug effect loses power almost identically. Under compound symmetry without carryover the joint distribution exists only for correlations below 0.256, so the published effect sizes of 0.3 and 0.6 relied on silently repaired matrices. Separable AR(1) correlation raises the bound to 0.44,

or 0.48 at a serial correlation of 0.7. A biomarker that is also prognostic makes analyses that share the biomarker effect with baseline anticonservative.

Conclusions. The covariance-based encoding is coherent but needs graded coupling, a correlation structure with room for realistic effects, and validity checks that stop instead of repairing. We give the closed-form results that identify the power loss, the code changes for the published software, and a refined process for simulation studies of aggregated N-of-1 designs.

Keywords: N-of-1 trials; predictive biomarker; simulation; carryover; data-generating process; power; prazosin.

Background

Aggregated N-of-1 trials and predictive biomarkers

In an aggregated N-of-1 trial each participant moves between drug and placebo, or between drug and no drug, several times, and the individual series are analyzed together[1–3]. Because every participant contributes both an on-drug and an off-drug response, the design can estimate how the drug effect varies between people, and so test whether a baseline measurement predicts who benefits[4,5]. The motivating example is prazosin for post-traumatic stress disorder (PTSD). Higher pretreatment standing systolic blood pressure was associated with greater symptom reduction on prazosin but not on placebo[6], which makes blood pressure a candidate predictive biomarker[7–9].

Closed-form power calculations exist only for simplified versions of these designs[10,11], so power is estimated by simulating trials from a data-generating process (DGP) and analyzing each simulated trial[12]. The DGP decides what a simulation can show. In particular, it decides how the biomarker interaction is represented, how the drug effect builds and wanes, and how carryover from one period reaches the next.

The framework of Hendrickson et al.

Hendrickson et al.[13] (hereafter RH) introduced the simulation framework this paper builds on. Its contributions include:

- **A component decomposition of the symptom score.** The score is the baseline level minus three responses, each following a modified Gompertz curve:
  - a biological drug response (BR), driven by time since the current on-drug period began;
  - an expectancy response (PB), driven by time in the trial and scaled by the participant's expectancy of receiving drug;
  - a time-varying natural course (TV).
- **Design machinery.** It covers open-label, crossover, blinded discontinuation and a Hybrid design combining all three. It tracks time on drug, time since discontinuation and expectancy for each visit.
- **Open software** (the `pmsimstats` R package).

RH also encoded the predictive biomarker in a distinctive way. Simulation studies of N-of-1 and crossover designs usually represent treatment-effect heterogeneity in one of two ways. The first is random participant-by-treatment variation[4,14–16]. The second, for an observed moderator, is a biomarker-by-treatment term in the mean[17–19]. RH instead placed the biomarker in the same multivariate normal distribution as the response components. They gave it a nonzero correlation c_{bm} with the drug response at on-drug visits and zero correlation elsewhere. A predictive biomarker is then one whose association with the outcome differs between on-drug and off-drug visits. To our knowledge this was the first time the biomarker-by-treatment interaction was parameterized as a covariance in the N-of-1 literature. The parameterization keeps the marginal distribution of the drug response fixed and expresses the strength of the interaction as a correlation.

The encoding is natural for a biomarker that tracks individual variation in response rather than a fixed dose-response relation, and it lets a single correlation calibrate the strength of the effect against the variability of the drug response.

Why a follow-up is needed

Reusing RH's process for a series of simulation studies exposed two problems. Both trace to the implementation, not to the idea.

1. **The joint distribution often does not exist.** Under the compound-symmetry structure RH used, the correlation matrix is positive definite only for c_{bm} below a ceiling, 0.256 in the

Hybrid design without carryover. RH's nonzero effect sizes were 0.3 and 0.6. The code repairs any non-positive-definite matrix silently. Counting one matrix per design path, effect size and half-life, 37% of the published grid's matrices at those effect sizes (20 of 54) were replaced by nearby valid matrices. At $c_{bm} = 0.3$ the repair changed no correlation by more than 0.007, so RH's results at that effect size are essentially those of the stated process. At $c_{bm} = 0.6$ it lowered the on-drug correlation to 0.54 to 0.57 and moved other correlations by up to 0.10, so those results describe a different process from the one stated.

2. **Power collapses under any carryover.** RH reported that power to detect the interaction fell sharply once carryover was introduced. In the Hybrid design at $N = 70$ and $c_{bm} = 0.3$ it fell from 0.74 to 0.12 at a half-life of 0.1 week, when only 0.1% of the drug effect remains a week after stopping. We show that this is an artifact of the coupling rule, not a property of the designs.

This paper therefore has three aims:

- to identify, in closed form, why power collapses under the published process;
- to give the changes to the published code that remove both problems;
- to specify a refined DGP, consistent with the pharmacology of short-half-life, titrated drugs such as prazosin, intuitive in its parameters and flexible in the features it can represent, for use across simulation studies of aggregated N-of-1 designs.

Methods

The published data-generating process

We use RH's notation, with a few additions. Participant i is observed at a baseline visit $t = 0$ and at post-baseline visits $t = 1, \dots, T$ (weeks 4, 8, 9, 10, 11, 12, 16 and 20 in the Hybrid design). The symptom score is

$$Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it}),$$

so a larger response lowers the score. Each component has a mean given by the modified Gompertz curve

$$G(x;\, m, d, r) = m\,\frac{\exp(-de^{-rx}) - e^{-d}}{1 - e^{-d}},$$

evaluated at a component-specific clock:

- **BR** at time on drug t_{od}, the time since the current on-drug period began, with $m = 10.99$, $d = 5$, $r = 0.42$ and SD 8. The curve reaches its inflection at 3.8 weeks, half its maximum at 4.7 weeks and 90% at 9.2 weeks.
- **PB** at time in the trial, scaled by the expectancy e_t (1 in the open-label phase, 0.5 when blinded), with $m = 6.51$, $d = 5$, $r = 0.35$ and SD $10e_t$.
- **TV** at time in the trial, with the same curve as PB and SD 10.

Off drug, the drug-response mean decays from its previous value with half-life $t_{1/2}$: $\mu_{BR,t} = \mu_{BR,t-1}\, 2^{-t_{sd,t}/t_{1/2}}$, where t_{sd} is the time since discontinuation.

The vector (biomarker, baseline, $TV_{i\cdot}$, $PB_{i\cdot}$, $BR_{i\cdot}$) is multivariate normal. Within each component the correlation between any two visits is $\rho = 0.8$ (compound symmetry). Between components it is 0.2 at the same visit and 0.1 otherwise. The biomarker's correlation with BR_{it} is $c_{bm}\, g_t$, where g_t is the *coupling*. RH's rule sets $g_t = 1$ wherever the drug-response mean is nonzero and 0 elsewhere.

RH analyzed each trial with the linear mixed model

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_D D_{it} + \beta_T t + \beta_{bm:D} B_i D_{it} + u_i + \varepsilon_{it},$$

with a random intercept u_i, the binary drug indicator D_{it}, and baseline included as a row with $D_{i0} = 0$. The interaction $\beta_{bm:D}$ is tested with Satterthwaite degrees of freedom[20]. Participants are assigned to the design's paths in fixed numbers (18, 18, 17 and 17 in the Hybrid design at $N = 70$), not at random.

Estimand

The target is the on-drug biomarker moderation of the drug response: the change in the drug effect per unit of biomarker while the drug is fully present. Without carryover it equals $\beta_{bm:D} = -c_{bm}\sigma_{BR}/\sigma_{bm}$ under covariance binding, and $-\beta_{bm}\sigma_{BR}/\sigma_{bm}$ under mean binding.

Analyses that contrast on-drug with off-drug visits through a binary drug indicator estimate this quantity only when no moderation remains off drug. Under carryover they estimate an attenuated on-minus-off contrast. Part of the power loss under graded coupling therefore reflects the binary indicator in the analysis, not only the data-generating process. Analyses with an exposure-decayed indicator $D_{bc,it}$ at the true half-life recover the on-drug moderation. The closed-form results and RH's analysis use the binary indicator; the stress test uses $D_{bc,it}$.

Closed-form analysis

To isolate how carryover acts, we use a summary statistic whose distribution can be derived exactly (Appendix A). For each participant,

$$\Delta_i = \bar{Y}_i^{\text{on}} - \bar{Y}_i^{\text{off}} = \sum_t a_t Y_{it},$$

the difference between the mean on-drug and mean off-drug scores, which is regressed on B_i by least squares. We call it E9 when it uses post-baseline visits only, and E9b when baseline is counted as one more off-drug visit, as in RH's analysis. Monte Carlo simulation of RH's mixed model shows that E9b reproduces its mean estimates to within 0.003 and its power closely (Results). The derivation gives three results.

- **Proposition 1.** The variance of Δ does not depend on the carryover half-life. The half-life enters only through the response means and the coupling, and neither appears in the variance.
- **Proposition 2.** The population slope is

$$\beta_{bm:D}^{\text{E9}} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\left(1-\bar{g}_{\text{off}}\right), \qquad \beta_{bm:D}^{\text{E9b}} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\left(1-\frac{n_{\text{off}}}{n_{\text{off}}+1}\,\bar{g}_{\text{off}}\right),$$

  where $\bar{g}_{\text{off}}$ is the mean coupling over the n_{off} post-baseline off-drug visits. The slopes are specific to each path: in the Hybrid design n_{off} is 3 in two paths and 4 in the other two. The pooled slope is their allocation-weighted average.
- **Power** is a noncentral-t probability governed by the correlation $\rho_{\Delta B}$ between Δ and B. For E9 that correlation is proportional to the *coupling gap* $1-\bar{g}_{\text{off}}$. For E9b it is proportional to $1-\frac{n_{\text{off}}}{n_{\text{off}}+1}\bar{g}_{\text{off}}$.

So carryover can affect power only through the coupling gap. The propositions reduce the question of why power collapses to the question of what the coupling rule does to $\bar{g}_{\text{off}}$.

The ceiling. Write the response block of the correlation matrix as M and the biomarker's correlations with the response components as $c_{bm}u$. The full matrix is positive definite if and only if M is, and

$$c_{bm}^2 \, u^\top M^{-1} u < 1.$$

The left side is the squared multiple correlation of the biomarker on the whole response trajectory, so the ceiling on c_{bm} is $(u^\top M^{-1} u)^{-1/2}$ (Appendix A). It depends on the correlation structure, the design and the coupling profile.

Refinements

Each refinement removes one defect (Table 1). The first three are corrections. The rest are modeling choices, and each can be switched off to recover the published process.

Table 1: Refinements to the published process.

Step	Change	Defect removed or feature added
1	Stop when the correlation matrix is not positive definite	Silent repair changes the correlations, and so the effect, without warning
2	Graded coupling: $g_t = 1$ on drug, $2^{-t_{sd,t}/t_{1/2}}$ off drug after exposure, 0 before any exposure	Step coupling removes the interaction at any positive half-life
3	Anchored carryover: decay the drug-response mean from its last on-drug value	The recursion compounds the decay at the second and later off-drug visits
4	AR(1) correlation within each component, $\rho^{\lvert t-s \rvert}$ in weeks	Compound symmetry overstates long-lag correlation
5	Separable cross-component structure, $c_1\rho^{\lvert t-s \rvert}$	Raises the ceiling; gives a valid factored generator
6	Options: prognostic biomarker couplings, phase-specific measurement error, decay form, dropout	Not representable in the published process

Why separability

Step 5 is more than a way to gain room for larger effects. Write the response block of the correlation matrix as M. Under the separable form $M = K \otimes A$, the product of a 3×3 component correlation K (1 on the diagonal, c_1 off it) and one AR(1) kernel $A_{ts} = \rho^{|w_t - w_s|}$ in calendar weeks. Four properties follow (Appendix A.5).

1. **Validity for every schedule.** The eigenvalues of M are products of those of K and A, so M is positive definite for any visit times. The non-separable AR(1) form used in earlier versions of this software adds a same-visit cross-component term that carries correlation without variance. It is positive definite only when the smallest eigenvalue of A exceeds $(c_1 - c_\times)/(1 - c_\times)$, which fails on dense schedules.
2. **RH's structure is already of this kind in pieces.** It is the sum of a person-level and an occasion-level separable term, each valid. That is why RH's matrices fail only through the biomarker coupling, never in the response block.
3. **An exact, interpretable ceiling.** The ceiling factors into a component term and a schedule term,

$$c^*_{bm} = \sqrt{1 - R^2}\,\left(u^\top A^{-1} u\right)^{-1/2}, \qquad R^2 = \frac{2c_1^2}{1 + c_1},$$

   where R^2 is the squared multiple correlation of the drug response on the other two components at one visit. The schedule term is a sum of per-step costs of the coupling pattern. A pattern that switches abruptly between closely spaced visits is expensive, which is why designs with many on-off switches can carry only small effects.
4. **It matches the analysis model.** Under separability the summed response is an AR(1) process up to the expectancy scaling of the placebo component, which is the residual structure that `corCAR1` assumes. Under the non-separable form it is AR(1) plus a white-noise term the analysis model omits. This is derived; its effect on test calibration has not been measured.

Pharmacological grounding

The refined process is specified for short-half-life, titrated drugs, with prazosin as the reference case.

- **Time scales.** Prazosin's plasma half-life is about 2.3 hours[21], against weekly visits. The plasma concentration therefore falls by orders of magnitude within a day of the last dose.

The symptom response may lag the concentration, so we allow a symptom-level half-life of 0.1 to 0.2 week (17 to 34 hours), about 7 to 15 plasma half-lives, as the pharmacological range. Longer carryover would be behavioral or psychological, to be simulated as a labeled sensitivity scenario.

- **Titration and onset.** The drug response follows the Gompertz curve in time on drug, which reflects dose titration and the accumulation of benefit. Its landmarks (half of maximum at 4.7 weeks) should be read clinically when the parameters are chosen.
- **Re-exposure.** RH's process restarts time on drug at zero when the drug is resumed. For prazosin this is justified, because the drug is re-titrated after an interruption. For drugs without titration, the response should instead rebuild from its residual level. This is an option, not a default.
- **Coupling.** Graded coupling makes the biomarker's association with the drug response fade at the rate the drug effect fades. That is the pharmacological reading of a predictive biomarker: it predicts the size of the drug effect, so it cannot predict a drug effect that is no longer present.

Covariance-based and mean-based binding

The biomarker can be bound to the drug effect in the covariance, as RH did, or in the mean. Under mean moderation, the drug response is shifted by $\beta_{bm}\, b_i\, \sigma_{BR}\, h_t$, where b_i is the standardized biomarker and h_t the exposure profile. Plain mean moderation uses $h_t = 1$ on drug and 0 off drug; decayed mean moderation uses $h_t = 2^{-t_{sd,t}/t_{1/2}}$ off drug, the profile of graded coupling. We compare the forms in the Discussion.

Reference specification

Table 2 collects the refined process. Values not listed are RH's.

Table 2: The refined data-generating process. Defaults are those used in the stress test.

Element	Specification	Default
Components	$Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$, Gompertz means as in RH	RH's parameters

Element	Specification	Default
Within-component correlation	AR(1) in weeks, $\rho^{\lvert t-s\rvert}$, one ρ for all three components (RH allowed one per component)	$\rho = 0.7$
Cross-component correlation	separable, $c_1\rho^{\lvert t-s\rvert}$; the Kronecker form applies to the correlation, since the PB SD scales with the expectancy	$c_1 = 0.2$
Path allocation	randomized to paths (RH: fixed numbers)	randomized
Biomarker coupling	c_{bm} on drug; $c_{bm}2^{-t_{sd}/t_{1/2}}$ off drug after exposure; 0 before exposure	$c_{bm} \leq$ ceiling
Carryover on the mean	decay from the last on-drug value, $\mu_{BR,\text{last}}\, 2^{-t_{sd}/t_{1/2}}$	$t_{1/2} = 0.1$ to 0.2 week (pharmacological)
Re-exposure	time on drug restarts at zero (re-titration)	on, for titrated drugs
Validity	stop if the correlation matrix is not positive definite	always
Prognostic couplings (option)	correlation of the biomarker with TV or PB; or mean moderation of TV, β_{bm}^{TV}	off; 0.3 in the stress test
Measurement error (option)	independent, SD by phase	off; 3 open label, 6 blinded in the stress test
Decay form (option)	exponential or Weibull with the same half-life	exponential
Dropout (option)	monotone, completely at random or at random given the last score	off; 20% in the stress test

Simulation study design

We designed and report the simulations following the ADEMP structure of Morris et al.[12]: aims, data-generating mechanisms, estimands, methods and performance measures. There are three studies (Table 3). The scripts are listed under Availability of data and materials.

Table 3: Design of the three simulation studies in the ADEMP structure. MCSE: Monte Carlo standard error of a rejection rate, $\sqrt{p(1-p)/R}$ for R replicates.

	1. Closed-form verification	2. The published code	3. Stress test of analyses
Aim	Check the closed-form moments and power of Appendix A, and the collapse under the step rule	Show on RH's own code that the coupling rule alone produces the collapse	Choose an analysis that holds its nominal size and keeps power under the refined process
Data-generating mechanisms	Covariance binding with step and graded coupling, and mean binding; compound symmetry and AR(1); Hybrid, crossover and OL+BDC designs; $N = 70$; grids of c_{bm} and $t_{1/2}$	RH's `generateData()` at commit 58b32a9, as published and with graded coupling only (Appendix B.1); $c_{bm} = 0.25$. (a) Hybrid, $N = 70$, no censoring, $t_{1/2} \in \{0, 0.1, 0.2, 1\}$; (b) four designs, $N \in \{35, 70\}$, $t_{1/2} \in \{0, 0.1, 0.2\}$, RH's censoring patterns	The refined process (Table 2); base cell separable AR(1), $\rho = 0.7$, $N = 70$, $t_{1/2} = 0.5$ week; 41 cells (15 null, 26 alternative) varying correlation structure, N, carryover, decay form, dropout, measurement error and prognostic pathways
Estimand	$\beta_{bm:D}$, and the mean of the paired-difference statistic	$\beta_{bm:D}$	$\beta_{bm:D}$
Methods	Paired-difference statistics E9 and E9b, closed form and simulated; RH's mixed model fitted with `lmer`	RH's own analysis pipeline (`lmer`, model-based test) on both versions	Fourteen analyses, all fitted to the same simulated trials and judged by the same decision rule; two-sided tests at $\alpha = 0.05$

	1. Closed-form verification	2. The published code	3. Stress test of analyses
Performance measures	Mean, SD and power of the statistic; agreement as (simulated − closed form)/MCSE	Power; mean interaction estimate; number of repaired correlation matrices	Null rejection rate; size-adjusted power and maximin shortfall; κ, the ratio of mean model SE to empirical SD; power-curve monotonicity, steepness and detectable coupling; failed fits
Replicates	5,000 per cell (MCSE of power at most 0.007); 500 to 1,000 for `lmer`	(a) 500 per cell (MCSE at most 0.022); (b) 250 per cell (at most 0.032)	1,000 per null cell (MCSE 0.0069 at a rate of 0.05), 500 per alternative cell (at most 0.022); 28,000 trials

Number of replicates. Replicate counts were set by the precision each comparison needs. In study 3, 1,000 null replicates give a two-MCSE band of 0.036 to 0.064 for a single cell, and pooling 11 null cells gives the band of 0.046 to 0.054 used in the Results; the pre-registered criterion also caps every null cell at 0.075. In study 1, a cell is flagged when the simulated and closed-form values differ by more than 2 MCSE, which chance alone produces in about 4.6% of cells.

Random numbers. Every cell, or chunk of a cell, sets its own seed from a fixed rule: a function of the cell's parameters in study 1, a shared seed per half-life (study 2a) or per design, sample size and half-life (study 2b), and $20261007 + 1000 \times \text{cell} + \text{chunk}$ in study 3. Results therefore do not depend on the number of cores or the order in which chunks run. Both versions of RH's code in study 2 use the same seeds, and all analyses in study 3 are fitted to the same trials, so the comparisons are paired. R's default Mersenne-Twister generator was used throughout.

Failed fits and repaired matrices. In study 3, a fit that returned an error was recorded as missing. Failures numbered 0 to 12 per analysis over 28,000 trials, and at most 0.6% in any cell, far below the pre-registered 2% flag. In study 2 the published pipeline returned an estimate for every replicate, and no correlation matrix was repaired in either version, because $c_{bm} = 0.25$ lies below the compound-symmetry ceiling of 0.256. Study 1 does not fit models for the paired-difference statistic,

so it has no failures.

Software. R 4.6.1, with lme4 2.0.1 and lmerTest 3.2.1 for RH's pipeline, nlme 3.1-169, mmrm 0.3.17, clubSandwich 0.7.0 and MASS 7.3-65.

Deviations from the pre-registration. The stress test was pre-registered, with 39 cells, 11 analyses and a fixed selection rule, in `docs/46` of the repository. Four changes were made after results were seen:

1. One amendment. The reference analysis (random intercept with continuous-time AR(1) residuals and CR2 standard errors) was replaced by the same model with its half-life chosen by AIC, and the power-symmetry screen was dropped because it tests a property that holds exactly by construction (Results).
2. Two cells with a prognostic association between the biomarker and baseline severity were added, bringing the total to 41 cells. They are judged with the other prognostic cells.
3. Three analyses were added to the eleven pre-registered ones and fitted to the same trials: the expectancy-term model with phase-specific residual variances, with CR2 and with model-based inference, and the expectancy-term model with its half-life chosen by AIC. All fourteen were judged by the same decision rule.
4. The pre-registration counted failed fits as non-rejections; the rejection rates reported here exclude them from the denominator. With the failure rates above, the two conventions differ by at most 0.0025 in any rejection rate.

Results

The published coupling rule removes the interaction

Under RH's rule every off-drug visit in the Hybrid design follows an exposure. With any positive half-life its drug-response mean is positive, so its coupling is 1. Hence $\bar{g}_{\text{off}} = 1$ and, by Proposition 2,

$$\beta^{\text{E9}}_{bm:D} = 0, \qquad \beta^{\text{E9b}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\frac{1}{n_{\text{off}}+1} \qquad \text{for every } t_{1/2} > 0.$$

Power is a step function of the half-life (Table 4, Figure 1). It takes its no-carryover value at

Table 4: Closed-form power by carryover half-life (weeks). Hybrid design, $N = 70$, $c_{bm} = \beta_{bm} = 0.25$, compound symmetry. E9: paired-difference statistic on post-baseline visits; E9b: the same with baseline counted as an off-drug visit, as in RH's analysis. Against Monte Carlo estimates (5,000 replicates per cell), 17 of 566 cell-variants differ by more than 2 Monte Carlo SE.

Coupling rule	Statistic	0	0.1	0.2	0.5	1	2
Step (published)	E9	0.721	0.050	0.050	0.050	0.050	0.050
	E9b	0.654	0.081	0.081	0.081	0.081	0.081
Graded	E9	0.721	0.721	0.712	0.631	0.473	0.278
	E9b	0.654	0.653	0.647	0.586	0.473	0.328
Mean moderation	E9	0.681	0.681	0.682	0.682	0.682	0.682
	E9b	0.619	0.619	0.620	0.620	0.621	0.621

$t_{1/2} = 0$ and a constant at every half-life above about 0.02 week, whether 0.1 week or 10. Below that, the recursive off-drug mean underflows to zero in double precision and the published rule switches coupling off at late off-drug visits. For E9b the constant is 0.081, a residue of the baseline row, which alone keeps zero coupling.

RH's analysis behaves the same way. In our reconstruction of the published process, the mean estimate falls from -0.126 to between -0.026 and -0.032 at every positive half-life, and power falls from 0.63 to between 0.08 and 0.11. In a separately seeded run of RH's own code (Table 5), power falls from 0.668 to 0.126 at 0.1 week. The difference between the two runs, about 2.4 combined Monte Carlo SE, is within the spread of the reconstruction's positive-half-life cells.

With constant coupling, the biomarker moderates the on-drug and off-drug responses equally, so there is no interaction left to detect. RH's finding that small carryover destroys power is therefore a property of the coupling rule, not of carryover or of the designs.

Graded coupling restores a smooth decline

Under graded coupling the mean off-drug coupling is

$$\bar{g}_{\text{off}}(t_{1/2}) = \frac{1}{n_{\text{off}}} \sum_{t\,\text{off}} 2^{-t_{sd,t}/t_{1/2}},$$

which is strictly increasing in $t_{1/2}$, because every off-drug visit has $t_{sd} \geq 1$ week. Two consequences follow.

- **Power is strictly decreasing in the half-life** for the path-stratified paired-difference statis-

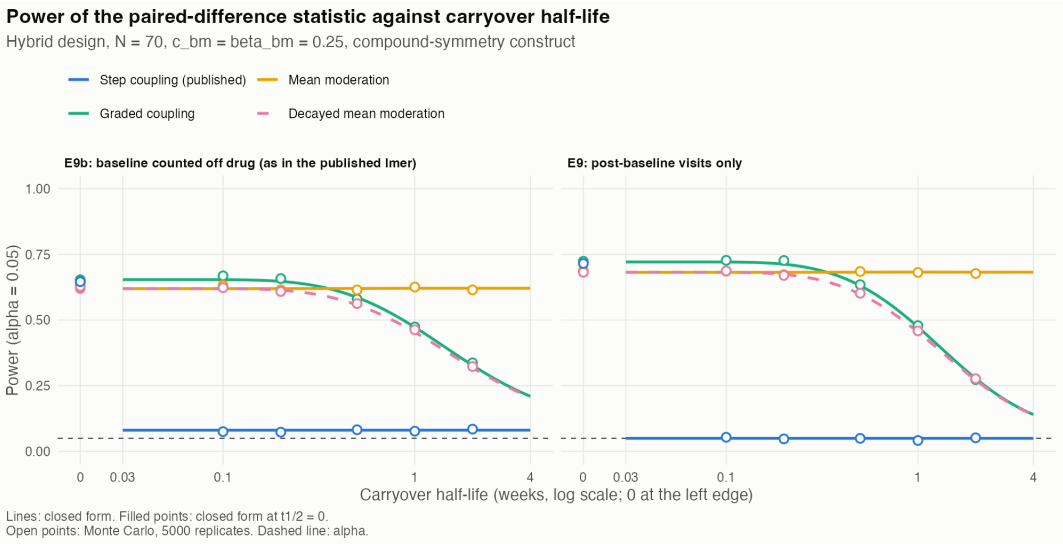


Figure 1: Closed-form power (lines) against carryover half-life, with Monte Carlo estimates (points), under the published step rule, graded coupling, mean moderation and decayed mean moderation. Hybrid design, $N = 70$, compound symmetry.

tic (Proposition 3, Appendix A). For the pooled statistic it is non-increasing at every step of a 120-point grid from 0.02 to 4 weeks. For RH's mixed model it was checked only at the simulated half-lives.

- **The loss vanishes faster than any power of the half-life as $t_{1/2} \to 0$.** At 0.1 week, $\bar{g}_{\text{off}} \leq 2^{-10}$ and power is unchanged to within 0.001.

The published rule is the limit of graded coupling as $t_{1/2} \to \infty$, applied at every positive half-life.

Power depends on the coupling profile only through the coupling gap $1-\bar{g}_{\text{off}}$ (Figure 2). The gap is 1 without carryover; under graded coupling it is 0.991 at 0.2 week, 0.906 at 0.5 and 0.754 at 1. Under the published rule it is 0 at every positive half-life. Because the noncentrality is approximately proportional to the gap when $\rho_{\Delta B}$ is small, the sample size needed for a given power scales roughly as $1/\text{gap}^2$.

The correction in the published code

Replacing only RH's coupling lines with graded coupling (Appendix B) and running their pipeline unchanged gave the following results (Table 5). The two versions coincide without carryover, as they must. At the published half-lives the graded version keeps the full interaction, and at 1 week it shows a gradual decline. No matrix was repaired in either version, because $c_{bm} = 0.25$ is below

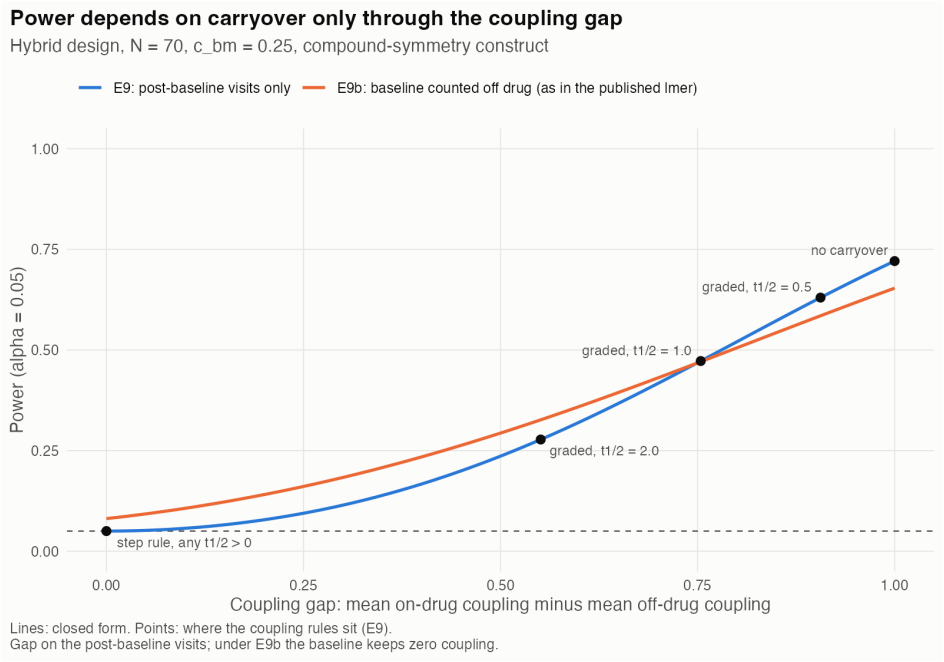


Figure 2: Closed-form power against the coupling gap, with the positions of the coupling rules. Hybrid design, $N = 70$, $c_{bm} = 0.25$.

Table 5: RH's code at commit 58b32a9, as published and with graded coupling: Hybrid design, $N = 70$, $c_{bm} = 0.25$, 500 replicates per cell, same seeds. Monte Carlo SE of power at most 0.022.

$t_{1/2}$ (weeks)	Mean estimate, published	Mean estimate, graded	Power, published	Power, graded
0	-0.131	-0.131	0.668	0.668
0.1	-0.028	-0.129	0.126	0.638
0.2	-0.029	-0.131	0.092	0.656
1	-0.033	-0.107	0.094	0.462

the ceiling.

Figure 3 repeats RH's Figure 4B on their own code, for all four designs, both sample sizes and the five censoring patterns, as published and with graded coupling. It uses $c_{bm} = 0.25$ instead of RH's 0.6, because 0.6 lies above every covariance-based ceiling and would confound the comparison with the repair. At 0.25 neither version repaired a matrix, so the coupling rule is the only difference. Results at $N = 70$ without censoring:

- **Hybrid design.** Published power falls from 0.66 to 0.08 at both positive half-lives. Graded power stays at 0.62 and 0.68.
- **Open label plus blinded discontinuation.** Published power falls from 0.57 to 0.12 and 0.13. Graded power stays at 0.57 and 0.60.

- **Crossover.** Published power falls only to 0.34 and 0.33, because the placebo-first path's off-drug visits precede any exposure and so keep zero coupling under both rules. Graded power stays at 0.71 and 0.78.
- **Open label.** There are no off-drug visits, so the two rules coincide and power is low throughout.

The same pattern holds at $N = 35$ and under every censoring pattern.

Figure 4 extends the comparison over interaction strengths and longer half-lives. It shows closed-form power in the Hybrid design for each coupling rule and for mean-based binding.

- **The published step rule.** Every column after $t_{1/2} = 0$ falls to the test size or just above it, at every strength. For example, at $c_{bm} = 0.2$ power under E9b falls from 0.46 to 0.07.
- **Graded coupling.** Power declines smoothly: 0.46, 0.46, 0.46, 0.41, 0.32 and 0.23 at the same strength over half-lives of 0 to 2 weeks.
- **Feasibility.** Gray cells mark effect sizes at or above the compound-symmetry ceiling, where no joint distribution exists. Under graded coupling, $c_{bm} = 0.3$ becomes feasible only from $t_{1/2} = 1$ week. Mean-based binding is feasible in every cell; it is flat in the half-life in its plain form, and declines like graded coupling when decayed.

Room for larger effect sizes

The ceiling $(u^\top M^{-1} u)^{-1/2}$ depends strongly on the correlation structure (Table 7).

- **Compound symmetry.** Under RH's structure the ceiling is 0.256 in the Hybrid design, below both of RH's nonzero effect sizes.
- **AR(1) within components,** with cross-component correlations that decay with the lag, raises the ceiling modestly, to 0.291 at $\rho = 0.8$. The effect depends on the design: in the open-label design, where every visit is on drug, AR(1) lowers the ceiling, because a single correlation repeated across visits that share little demands more of the biomarker.
- **The separable form** makes the cross-component correlation follow the same AR(1) decay ($c_1 \rho^{|t-s|}$), so the response block is a Kronecker product. The ceiling rises to 0.440 at RH's $\rho = 0.8$ and to 0.474 at $\rho = 0.7$ (0.480 in the Hybrid design).

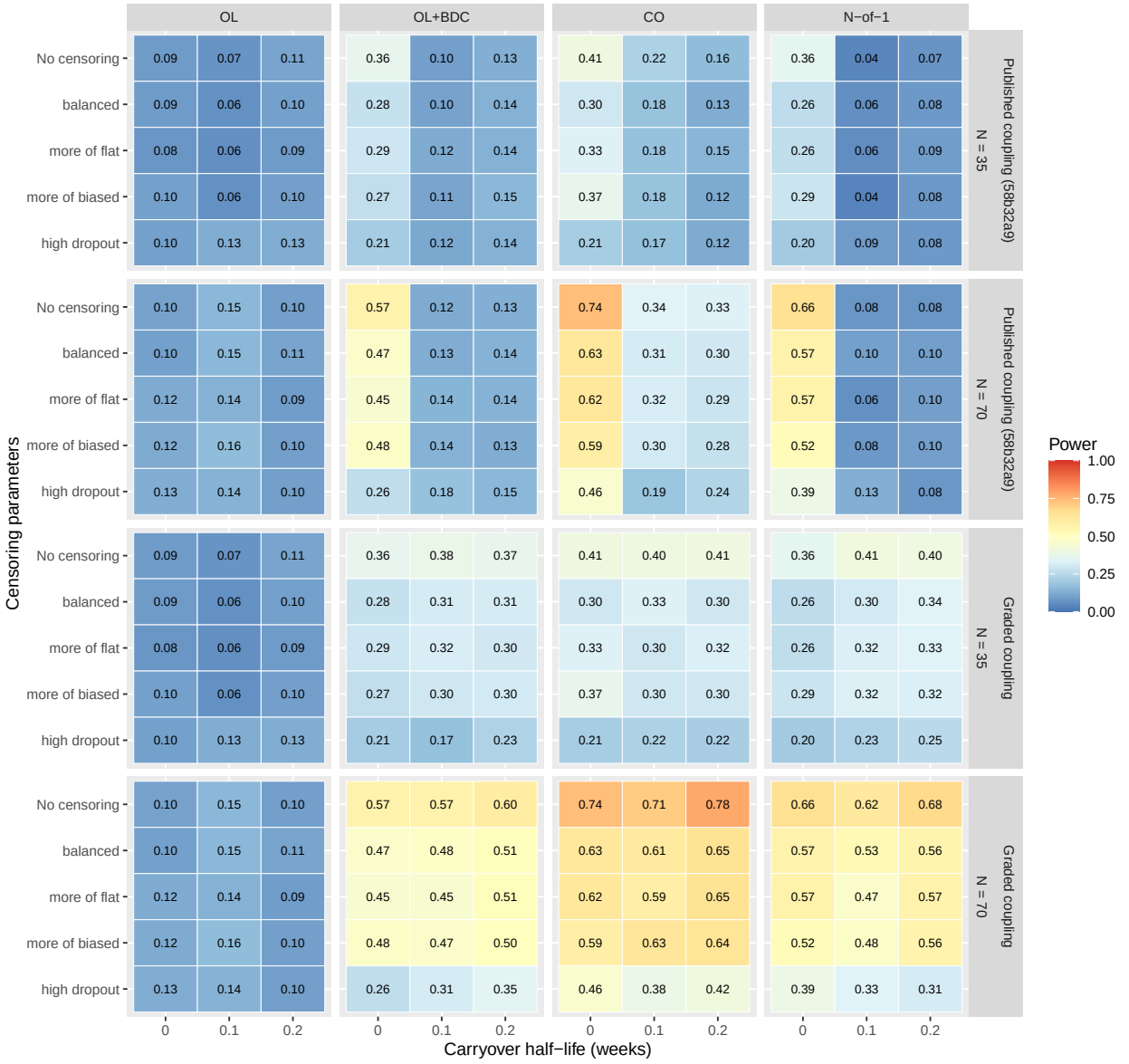


Figure 3: Power to detect the biomarker-by-treatment interaction, in the layout of RH's Figure 4B, from RH's code at commit 58b32a9 as published (upper two panel rows) and with graded coupling (lower two panel rows). Rows within panels: censoring patterns; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, 250 replicates per cell (Monte Carlo SE at most 0.032), the same seeds for both versions; no matrix was repaired in either version.

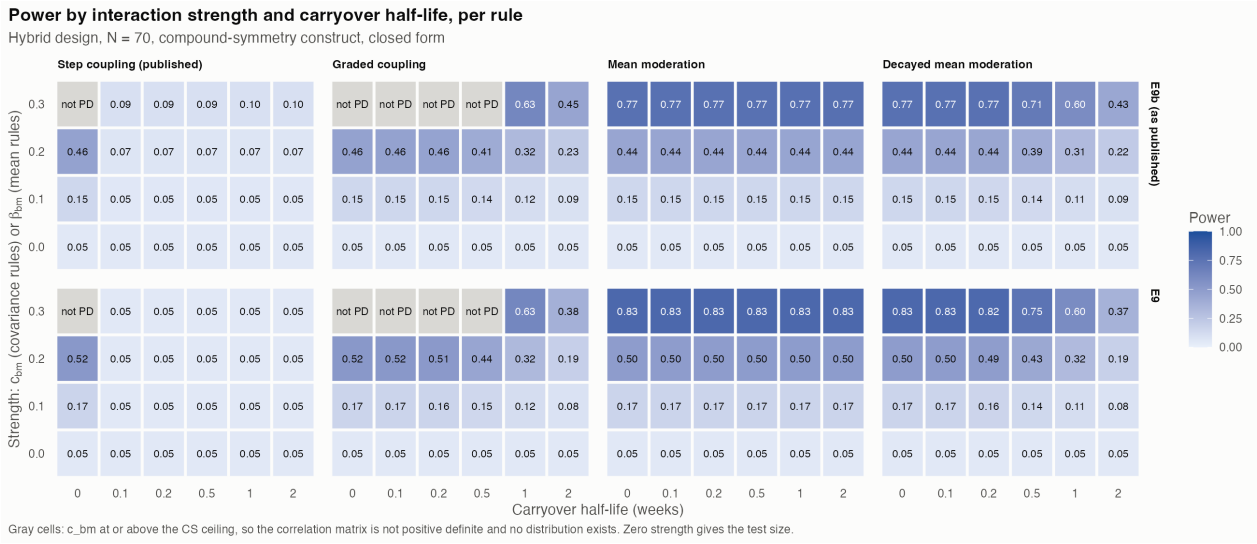


Figure 4: Closed-form power of the paired-difference statistic by interaction strength (c_{bm} for the covariance rules, β_{bm} for the mean rules) and carryover half-life, in the layout of RH's Figure 4B: published step coupling, graded coupling, mean moderation and decayed mean moderation. Upper row: E9b, which counts baseline as an off-drug visit as RH's analysis does; lower row: E9. Gray cells: c_{bm} at or above the compound-symmetry ceiling, where the correlation matrix is not positive definite. Hybrid design, $N = 70$, compound symmetry.

The higher ceiling has a cost. At the same c_{bm}, AR(1) data carry less information about the interaction than compound-symmetric data: the closed-form power of the paired-difference statistic in the Hybrid design at $c_{bm} = 0.25$ falls from 0.721 to 0.335. A larger feasible correlation does not by itself mean more power, so effect sizes under the two structures should be compared on the scale of the implied slope, not of c_{bm}.

Table 6 counts the correlation matrices that fail the published code's own positive-definiteness test (`corpcor::is.positive.definite()`), which would trigger its silent repair. There is one matrix per design path, effect size and half-life; N does not enter the matrix. Two points stand out.

- **At $c_{bm} = 0.3$ the refined process needs no repair, whereas the published process repairs 30% of its matrices.**
- **At $c_{bm} = 0.6$ the refined process fails more often than the published one.** The published process passes those cells at positive half-lives only because its step rule makes the coupling constant after first exposure. A constant coupling is cheap to accommodate, and it carries no interaction. The cells that stop needing repair under the published rule are the cells that have lost their signal.

Table 6: Correlation matrices that are not positive definite, $\rho = 0.8$. Published grid: open label, open label plus blinded discontinuation, crossover and Hybrid designs; $t_{1/2} \in \{0, 0.1, 0.2\}$ week. Extended grid: the last three designs; $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$ week. The published-process counts reproduce those of RH's own code exactly.

Grid (number of matrices)	Published (step coupling, compound symmetry)	Refined (graded coupling, separable AR(1))	Mean-based binding
Published Figure 4 grid, $c_{bm} \in \{0, 0.3, 0.6\}$ (81)	20 (25%)	21 (26%)	0 (0%)
Extended grid, $c_{bm} = 0.25$ (40)	0 (0%)	0 (0%)	0 (0%)
Extended grid, $c_{bm} = 0.3$ (40)	12 (30%)	0 (0%)	0 (0%)
Extended grid, $c_{bm} = 0.6$ (40)	12 (30%)	30 (75%)	0 (0%)

Table 7: Largest feasible biomarker correlation c_{bm} under graded coupling, without carryover, minimum over the Hybrid, crossover and open-label plus blinded discontinuation designs. Under the separable form at $\rho = 0.7$ the Hybrid value is 0.480.

ρ	Compound symmetry	AR(1), decaying cross-correlation	Separable AR(1)
0.8	0.256	0.291	0.440
0.7	0.346	0.452	0.474

Only mean-based binding represents every cell.

A biomarker correlation of 0.45 is feasible in every design under the separable form at $\rho = 0.7$, and not under the published structure. Effects as large as RH's 0.6 exceed every covariance-based ceiling in the Hybrid design. They can be represented only by binding the biomarker in the mean.

Consequences for the analysis

The refined process can simulate features the published one cannot, and these change which analysis is safe. The first result below bears on the DGP directly; the second concerns the analyses.

A prognostic biomarker. Suppose blood pressure is associated with the natural course or with placebo response as well as with the drug effect. We simulated a correlation of 0.3 with the natural-course component (constant over visits, or growing along its curve), with the placebo component, or with baseline severity. Analyses that keep baseline as an outcome row use one biomarker main effect β_B for baseline and for the off-drug visits. When the biomarker is prognostic, those two associations differ, and part of the difference is attributed to the interaction (Appendix A.4).

Every analysis that shares the biomarker effect with baseline failed in at least one prognostic cell:

Analysis	Natural course, constant	Natural course, growing	Placebo response	Baseline severity
Published (random intercept, model-based)	0.107	0.053	0.159	0.075
Random intercept + CAR(1), CR2	0.099	0.049	0.100	0.050
The same, half-life chosen by AIC	0.115	0.058	0.112	0.055
Unstructured covariance, Kenward-Roger	0.416	0.195	0.222	0.051
The same, half-life chosen by AIC	0.509	0.245	0.256	0.064
Random intercept + CAR(1) with expectancy terms, CR2	0.047	0.046	0.040	0.040
The same, with phase-specific residual variances, CR2	0.055	0.044	0.042	0.042
The same, with phase-specific residual variances, model-based	0.062	0.043	0.045	0.046

Analysis	Natural course, constant	Natural course, growing	Placebo response	Baseline severity
Expectancy terms, half-life chosen by AIC, CR2	0.059	0.052	0.047	0.045
Random intercept + CAR(1) with free baseline slope, CR2	0.049	0.063	0.049	0.050

Null rejection rates, 1,000 replicates per cell.

The analyses that free the biomarker's baseline association held the nominal level in every prognostic cell. One family adds expectancy and a biomarker-by-expectancy term to the mean model, with or without residual variances that differ by trial phase; the other adds `bmc:bl0`, which gives baseline its own biomarker slope. A prognostic association with baseline severity alone did little harm: it shifts the baseline row and the later visits equally.

Size more generally. Pooled over 11 non-prognostic null cells (11,000 replicates; nominal band 0.046 to 0.054):

- the published analysis rejected 0.069, and 0.081 at $N = 35$;
- an unstructured model with Kenward-Roger inference[22] rejected 0.057, with 0.087 at $N = 35$;
- random-intercept models with continuous-time AR(1) residuals and CR2 cluster-robust standard errors[23,24] rejected 0.048 to 0.053, with no single cell above 0.069; with phase-specific residual variances and model-based inference the rejection rate was 0.047;
- the expectancy-term model with its half-life chosen by AIC rejected 0.057, with 0.073 when the true half-life was one week, and so failed the size criterion.

Fourteen analyses were judged by the decision rule. Five passed the size, prognostic and sensitivity screens: the expectancy-term model, its two versions with phase-specific residual variances, and

the two models that free the baseline biomarker slope. Among them, the largest shortfall in size-adjusted power from the best nominal analysis, over the non-prognostic alternative cells, was 0.242 for the expectancy-term model, 0.252 and 0.232 for its phase-variance versions with CR2 and model-based inference, and 0.278 and 0.318 for the free-baseline models. The expectancy-term model and its model-based phase-variance version lie within the pre-registered tie margin of 0.02. Neither reaches a power of 0.80 on the grid, so the tie cannot be broken by detectable coupling, and the rule selects the expectancy-term model as the one with fewer covariance parameters. The phase-variance versions are effectively equivalent alternatives. One screen, for power symmetry in the sign of the coupling, proved defective: it tested a property that holds exactly by construction, so its failures were Monte Carlo error, and it was dropped. The power cost of robustness to a prognostic biomarker is about a third under AR(1) data and small under compound symmetry.

A simulation that omits a prognostic pathway cannot reveal these failures. The refined process includes it as an option for that reason.

Discussion

Covariance-based and mean-based binding

What the two encodings share. Under joint normality, conditioning on the biomarker shows how the covariance encoding relates to mean moderation. With b_i the standardized biomarker,

$$E(BR_{it} \mid B_i) = \mu_{BR,t} + c_{bm}\, g_t\, \sigma_{BR}\, b_i, \qquad \mathrm{Cov}(BR_{i\cdot} \mid B_i) = \Sigma_{BR} - c_{bm}^2 \sigma_{BR}^2\, g g^\top.$$

The covariance encoding is therefore mean moderation with exposure profile equal to the coupling profile g_t and strength $\beta_{bm} = c_{bm}$. It differs only in that its residual drug-response covariance is reduced by a rank-one term. Three consequences follow.

- **The power loss under carryover is set by the profile.** Mean moderation with the same fading profile loses power almost identically (below), and this is a consequence of the identity, not an independent finding.
- **The small power advantage of the covariance encoding** (0.721 against 0.681 for E9 without carryover) comes from the rank-one reduction of the residual variance. It is a matter

of how the variance is parameterized, not of the encoding.

- **The ceiling is the condition** that the reduced residual covariance remain positive definite.

RH's contribution is therefore a parameterization of the interaction, as a correlation that keeps the marginal variance of the drug response fixed, rather than a different model of the mean. That parameterization is what was new in the N-of-1 literature, to our knowledge, and both its strengths and its defects follow from it.

RH's covariance-based encoding has real strengths.

- **One interpretable parameter.** The interaction strength is a single correlation; its square, c_{bm}^2, is the share of on-drug drug-response variance the biomarker explains.
- **A reduced form of responder heterogeneity.** If participants fall into latent responder classes, and both the biomarker and the drug response differ between classes, then the covariance between them is carried by class membership. Matching second moments gives $c_{bm}^2 = f_B f_{BR}$, where f_B and f_{BR} are the fractions of the biomarker's and the drug response's variance that lie between classes[25]. The encoding therefore matches the second moments of a responder-class mechanism without requiring the mixture to be identified, which typically needs larger samples than N-of-1 programs enroll[25]. The match is limited to second moments: the mixture's conditional mean is nonlinear in the biomarker, and its higher moments differ.
- **The marginal variance of the drug response is fixed.** The biomarker explains part of the existing variation rather than adding to it, so the effect size can be stated as a share of response variance without changing the response distribution.
- **It fits a biomarker that marks individual sensitivity to the drug** better than one that sets a fixed dose-response shift.

Its weaknesses, all made explicit by this analysis, are structural.

- **The joint distribution must exist,** which bounds the effect size below values that are clinically plausible. The bound depends on the correlation structure and the design.
- **The coupling must be specified at off-drug visits,** and the natural choice of an indicator is wrong under carryover.
- **The drug-response component keeps its full variance at off-drug visits,** with zero mean. This adds noise to every on-off contrast. It belongs to the component construct, not to covariance binding: RH's mean-moderation variant shares it.

- **A correlation is less intuitive than a mean effect for clinical readers.** The slope it implies, $-c_{bm}\sigma_{BR}/\sigma_{bm}$, must be derived before it can be compared with an observed effect. Even then the comparison needs a common outcome scale. The reported association for prazosin, 14 points on the Clinician-Administered PTSD Scale per 10 mm Hg of standing systolic pressure[6], is on a scale the simulated symptom score does not share. Calibrating RH's parameters to such data is future work.

Mean moderation has no ceiling, an intuitive parameter (points of symptom change per SD of biomarker) and no coupling to specify off drug. In its plain form the shift is applied only at on-drug visits, so its power is flat in the half-life (0.681 at every value for E9). But it adds variance instead of explaining it, and that flatness is a coding choice, not a finding: the biomarker's effect stops the moment the drug is stopped, while the drug effect itself carries over.

When the mean shift is decayed with the drug effect, as graded coupling decays the correlation, power declines almost exactly as it does under graded coupling: 0.681, 0.681, 0.674, 0.598, 0.454 and 0.271 at half-lives of 0, 0.1, 0.2, 0.5, 1 and 2 weeks, against 0.721, 0.721, 0.712, 0.631, 0.473 and 0.278. As the identity above implies, how much carryover costs is decided by the exposure profile of the interaction, whether it fades with the drug effect, and not by whether it is bound in the covariance or in the mean. Comparisons that report larger carryover losses under covariance binding than under mean binding are comparing a fading interaction with a non-fading one.

A natural next step combines the strengths. Generate the drug response from a participant-level sensitivity to exposure,

$$BR_{it} = \omega_i f(t) + \varepsilon_{it}, \qquad \omega_i = \bar{\omega}\,(1 + \zeta b_i) + u_i,$$

where $f(t)$ is the exposure-response time course, ζ the biomarker's effect on sensitivity and u_i the sensitivity it does not explain. This is a random-coefficient model with a covariate-by-exposure interaction, the form used to describe individual response in replicated crossover and N-of-1 trials[4,14].

- It binds the biomarker in the mean through individual sensitivity, so it has no ceiling.
- Its moderation is proportional to the response and fades with it.
- Its off-drug variance shrinks with the response.

It is not yet implemented in the software, and its response half-life for prazosin in PTSD is not known.

Recommendations

For simulation studies that build on RH, we recommend the following.

1. **Always apply the corrections:** stop on non-positive-definite matrices, use graded coupling, and use anchored carryover.
2. **Use the separable AR(1) structure** at a stated ρ, and choose c_{bm} below the ceiling for every design in the study.
3. **Keep RH's process as a reference arm** whenever results are compared with the published ones. Attribute each difference to a named refinement.
4. **Report effect sizes on two scales:** as c_{bm}, and as the implied slope in outcome points per SD of biomarker.
5. **Include a prognostic scenario** whenever the analysis shares the biomarker effect with the baseline visit.
6. **Use mean moderation, or the sensitivity model once available, for effects above the covariance ceiling,** and as a sensitivity analysis on the encoding itself.

Limitations

- **The closed form covers one statistic.** It is exact for the paired-difference statistic, not for RH's mixed model. The two agree closely in the Hybrid design, but in the crossover design the closed form understates what a time-adjusted analysis achieves.
- **Parameter values are not estimated.** The Gompertz parameters, correlations and prognostic couplings are RH's values or chosen values, not estimates from prazosin trial data.
- **The process is Gaussian with a single biomarker.**
- **The pharmacological carryover range** for prazosin at weekly visits is inferred from its plasma half-life, not measured as a symptom-response half-life.

Conclusions

RH introduced a covariance-based encoding of the predictive-biomarker interaction that remains a coherent choice for simulating aggregated N-of-1 trials. Its published implementation removes the interaction whenever there is carryover, and it silently alters correlations it cannot represent. Both problems have closed-form explanations and small fixes. With graded coupling, a separable correlation structure and validity checks that stop instead of repairing, the process behaves as the pharmacology of a short-half-life drug suggests it should: power declines smoothly with the half-life, and negligibly at pharmacological half-lives. We propose the refined process, with options for prognostic pathways and other features that change which analysis is valid, as the reference for simulation studies of these designs, pending calibration of its parameters to trial data.

Abbreviations

BR: biological drug response; CO: crossover; CR2: bias-reduced cluster-robust variance estimator; CS: compound symmetry; DGP: data-generating process; OL+BDC: open label followed by blinded discontinuation; PB: expectancy (placebo-belief) response; PTSD: post-traumatic stress disorder; RH: Hendrickson et al. (2020); TV: time-varying natural course.

Declarations

Ethics approval and consent to participate. Not applicable: the study uses simulated data only.

Consent for publication. Not applicable.

Availability of data and materials. All code and simulated data are in the pmsimstats-ng repository:

- the closed-form calculations, in `analysis/scripts/quick-sim/carryover-closed-form/01-e9-closed-fo`
- the check of the published code, in `analysis/scripts/quick-sim/hendrickson-problems/14-graded-patc`
- the stress test, in `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R` and `12-strawman-stress-decision.R`;

- the published code, at `github.com/rchendrickson/pmsimstats`, commit 58b32a9.

Competing interests. To be completed by the authors. Any author of this paper who is also an author of RH should be disclosed here.

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References

1. Nikles J, Mitchell GK, Schluter P, et al. Aggregating single patient (n-of-1) trials in populations where recruitment and retention was difficult: The case of palliative care. *Journal of Clinical Epidemiology*. 2011;64(5):471-480. doi:10.1016/j.jclinepi.2010.05.009

2. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 Statement. *Journal of Clinical Epidemiology*. 2016;76:9-17. doi:10.1016/j.jclinepi.2015.05.004

3. Punja S, Xu D, Schmid CH, et al. N-of-1 trials can be aggregated to generate group mean treatment effects: A systematic review and meta-analysis. *Journal of Clinical Epidemiology*. 2016;76:65-75. doi:10.1016/j.jclinepi.2016.03.026

4. Senn S. Mastering variation: Variance components and personalised medicine. *Statistics in Medicine*. 2016;35(7):966-977. doi:10.1002/sim.6739

5. Schork NJ. Personalized medicine: Time for one-person trials. *Nature*. 2015;520(7549):609-611. doi:10.1038/520609a

6. Raskind MA, Millard SP, Petrie EC, et al. Higher pretreatment blood pressure is associated with greater posttraumatic stress disorder symptom reduction in soldiers treated with prazosin. *Biological Psychiatry*. 2016;80(10):736-742. doi:10.1016/j.biopsych.2016.03.2108

526 7. Raskind MA, Peskind ER, Kanter ED, et al. Reduction of nightmares and other PTSD symptoms in combat veterans by prazosin: A placebo-controlled study. *American Journal of Psychiatry*. 2003;160(2):371-373.

527 8. Raskind MA, Peterson K, Williams T, et al. A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from iraq and afghanistan. *American Journal of Psychiatry*. 2013;170(9):1003-1010.

528 9. Raskind MA, Peskind ER, Chow B, et al. Trial of prazosin for post-traumatic stress disorder in military veterans. *New England Journal of Medicine*. 2018;378(6):507-517. doi:10.1056/NEJMoa1507598

529 10. Wang Y, Schork NJ. Power and design issues in crossover-based N-of-1 clinical trials with fixed data collection periods. *Healthcare (Basel)*. 2019;7(3):84. doi:10.3390/healthcare7030084

530 11. Senn S. *Cross-over Trials in Clinical Research*. 2nd ed. John Wiley & Sons; 2002.

531 12. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086

532 13. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and theoretical findings. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013

533 14. Araujo A, Julious S, Senn S. Understanding variation in sets of N-of-1 trials. *PLoS ONE*. 2016;11(12):e0167167. doi:10.1371/journal.pone.0167167

534 15. Senn S. Sample size considerations for n-of-1 trials. *Statistical Methods in Medical Research*. 2019;28(2):372-383. doi:10.1177/0962280217726801

535 16. Zucker DR, Schmid CH, McIntosh MW, D'Agostino RB, Selker HP, Lau J. Combining single patient (N-of-1) trials to estimate population treatment effects and to evaluate individual patient responses to treatment. *Journal of Clinical Epidemiology*. 1997;50(4):401-410. doi:10.1016/S0895-4356(96)00429-5

536 17. Haller B, Ulm K. A simulation study on estimating biomarker-treatment interaction effects in randomized trials with prognostic variables. *Trials*. 2018;19(1):128.

537 18. Grenet G, Blanc C, Bardel C, Gueyffier F, Roy P. Comparison of crossover and parallel-group designs for the identification of a binary predictive biomarker of the treatment effect. *Basic and Clinical Pharmacology and Toxicology*. 2020;126(1):59-64. doi:10.1111/bcpt.13293

538 19. Kent DM, Paulus JK, Klaveren D van, et al. The predictive approaches to treatment effect heterogeneity (PATH) statement. *Annals of Internal Medicine*. 2020;172(1):35-45. doi:10.7326/M18-3667

539 20. Kuznetsova A, Brockhoff PB, Christensen RHB. lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software*. 2017;82(13):1-26. doi:10.18637/jss.v082.i13

540 21. Hobbs DC, Twomey TM, Palmer RF. Pharmacokinetics of prazosin in man. *Journal of Clinical Pharmacology*. 1978;18(8-9):402-406. doi:10.1002/j.1552-4604.1978.tb02456.x

541 22. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*. 1997;53(3):983-997. doi:10.2307/2533558

542 23. Bell RM, McCaffrey DF. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*. 2002;28(2):169-181.

543 24. Pustejovsky JE, Tipton E. Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. *Journal of Business and Economic Statistics*. 2018;36(4):672-683. doi:10.1080/07350015.2016.1247004

544 25. pmsimstats team. Latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials. Published online 2026.

545 26. Liu GF, Lu K, Mogg R, Mallick M, Mehrotra DV. Should baseline be a covariate or dependent variable in analyses of change from baseline in clinical trials? *Statistics in Medicine*. 2009;28(20):2509-2530. doi:10.1002/sim.3639

Appendix A. Closed-form derivations

A.1 Moments of the paired-difference statistic

Let $a_t = 1/n_{\text{on}}$ at on-drug visits and $-1/n_{\text{off}}$ at off-drug visits, so that $\Delta_i = \sum_t a_t Y_{it}$. Because the weights sum to zero, the baseline level cancels and $\Delta = -\sum_t a_t S_t$, with $S_t = TV_t + PB_t + BR_t$.

With $m_1 = \sum a_t$, $m_2 = \sum a_t^2$, $m_e = \sum a_t e_t$, $m_{ee} = \sum a_t^2 e_t^2$ and $m_{2e} = \sum a_t^2 e_t$, the variance under compound symmetry is

$$\begin{aligned}\operatorname{Var}(\Delta) = {} & (\sigma_{TV}^2 + \sigma_{BR}^2)[\rho m_1^2 + (1-\rho)m_2] + \sigma_{PB}^2[\rho m_e^2 + (1-\rho)m_{ee}] \\ & + 2\sigma_{TV}\sigma_{BR}[c_\times m_1^2 + (c_1 - c_\times)m_2] + 2(\sigma_{TV} + \sigma_{BR})\sigma_{PB}[c_\times m_1 m_e + (c_1 - c_\times)m_{2e}],\end{aligned}$$

where c_1 and $c_\times$ are the same-visit and other-visit cross-component correlations.

- **Proposition 1.** No term depends on $t_{1/2}$, so $\operatorname{Var}(\Delta)$ is free of the half-life. The argument applies to any construct in which the half-life does not enter the response block. It therefore holds for unrepaired matrices only: the repair changes the response block, and its changes depend on the half-life through the coupling. For E9 the weights sum to zero, so $m_1 = 0$. For E9b the post-baseline weights sum to $1/(n_{\text{off}} + 1)$, and the baseline row, with $S_0 = 0$, carries the remaining weight.
- **Proposition 2.** Only BR_t is correlated with B, with covariance $c_{bm} g_t \sigma_{bm} \sigma_{BR}$, so $\beta_{bm:D} = -c_{bm}(\sigma_{BR}/\sigma_{bm}) \sum_t a_t g_t$. With $g_t = 1$ on drug, this gives the expressions in the Methods.
- **Residual variance.** Under covariance binding, $\tau^2 = \operatorname{Var}(\Delta) - \beta_{bm:D}^2 \sigma_{bm}^2$. Under mean binding the shift adds variance and $\tau^2 = \operatorname{Var}(\Delta)$.

A.2 Power

Within a path of n participants, the least-squares slope is exactly normal given the biomarker values. Its t-statistic is exactly noncentral t with $n-2$ degrees of freedom and noncentrality $\beta_{bm:D}\sqrt{\text{SS}_B}/\tau$, where $\text{SS}_B \sim \sigma_{bm}^2 \chi_{n-1}^2$.

At $\text{SS}_B = \sigma_{bm}^2(n-1)$ the noncentrality is $\Lambda = \sqrt{n-1}\,\rho_{\Delta B}/\sqrt{1-\rho_{\Delta B}^2}$, where $\rho_{\Delta B} = \beta_{bm:D}\sigma_{bm}/\text{SD}(\Delta)$ is proportional to the coupling gap.

Pooling paths with a common intercept adds the between-path variation in mean contrast to the residual variance; in the Hybrid design it is under 0.7% of the total. Power is the noncentral-t tail probability averaged over the χ^2 law of SS_B.

A.3 Monotonicity under graded coupling

$$\frac{d\bar{g}_{\text{off}}}{dt_{1/2}} = \frac{\ln 2}{n_{\text{off}}\, t_{1/2}^2} \sum_{t\,\text{off}} t_{sd,t}\, 2^{-t_{sd,t}/t_{1/2}} > 0.$$

- **Proposition 3.** By Proposition 2 the slope's magnitude is strictly decreasing in $t_{1/2}$. By A.1, τ^2 is strictly increasing, since $\mathrm{Var}(\Delta)$ is fixed. So the noncentrality and the power are strictly decreasing.
- **Small half-lives.** $\bar{g}_{\text{off}} \le 2^{-1/t_{1/2}}$, which gives the behavior near zero stated in the Results.
- **Large half-lives.** $\bar{g}_{\text{off}} \to 1$ as $t_{1/2} \to \infty$, which recovers the step rule.

A.4 The ceiling and the shared biomarker effect

The ceiling. Order the variables as biomarker, then responses, and partition the correlation matrix as $\begin{pmatrix} 1 & c_{bm}u^\top \\ c_{bm}u & M \end{pmatrix}$. By the Schur complement, the matrix is positive definite if and only if M is positive definite and $1 - c_{bm}^2 u^\top M^{-1} u > 0$. Hence the ceiling is $c_{bm}^* = (u^\top M^{-1} u)^{-1/2}$.

The shared biomarker effect. Let s_0, s_{off} and s_{on} be the true slopes of Y on B at baseline, at post-baseline off-drug visits and at on-drug visits. A model with baseline as an off-drug row estimates one β_B for s_0 and s_{off}.

The following account is heuristic, not a derivation. Consider a prognostic biomarker with no interaction, whose post-baseline slope is constant: $s_0 = 0 \neq s_{\text{off}} = s_{\text{on}} = s_{\text{prog}}$.

- **The leak.** The fitted $\hat{\beta}_B$ lies between the baseline and off-drug slopes, $\hat{\beta}_B \approx f\, s_{\text{prog}}$ for some $0 < f < 1$ that depends on the weight the baseline row receives. The interaction absorbs the remainder, $\hat{\beta}_{bm:D} \approx (1 - f)\, s_{\text{prog}} \neq 0$.
- **The remedy.** Adding $B_i \cdot \mathbf{1}\{t = 0\}$ frees s_0 and removes this source of bias. That the baseline row imposes such a constancy constraint is known from the literature on baseline as a covariate or a response[26].

Two further mechanisms operate in the stress-test scenarios.

- **The slope grows over time.** When the post-baseline slope grows, the bias depends on how drug status is confounded with time. Freeing s_0 is then not sufficient in general. The residual 0.063 for that analysis is consistent with this.
- **Expectancy.** When the biomarker is associated with the placebo response, the slope also scales with the expectancy e_t, which is higher at the open-label (on-drug) visits. The expectancy term $B_i \cdot e_t$, with $e_0 = 0$, addresses both this mechanism and the baseline constraint.

A scenario in which the biomarker is correlated with baseline severity was not simulated.

A.5 The separable structure

Let $A_{ts} = \rho^{|w_t - w_s|}$ and $K = (1 - c_1)I_3 + c_1 J_3$, and set $M = K \otimes A$, so that the correlation between component c at visit t and component c' at visit s is $K_{cc'}A_{ts}$.

Validity. The eigenvalues of M are the products of those of K and A.

- K has eigenvalues $1 + 2c_1$ and $1 - c_1$ (twice), so it is positive definite for $-\frac{1}{2} < c_1 < 1$.
- A is the correlation matrix of a stationary Ornstein-Uhlenbeck process at the visit times, and is positive definite for any distinct times.

Hence M is positive definite for every schedule. Generatively, $\mathbf{X} = L_A \mathbf{Z} L_K^\top$ with $L_A L_A^\top = A$ and $L_K L_K^\top = K$: each column of $L_A \mathbf{Z}$ is an independent AR(1) path, and the components are fixed mixtures of them.

The ceiling. With the biomarker coupled to the drug-response component only, its correlation vector is $c_{bm}(e_3 \otimes u)$. Since $M^{-1} = K^{-1} \otimes A^{-1}$,

$$(e_3 \otimes u)^\top M^{-1}(e_3 \otimes u) = [K^{-1}]_{33}\, u^\top A^{-1} u, \qquad [K^{-1}]_{33} = \frac{1 + c_1}{(1 - c_1)(1 + 2c_1)} = \frac{1}{1 - R^2}.$$

The schedule term. The sampled process is Markov, with $\phi_t = \rho^{w_t - w_{t-1}}$, so

$$u^\top A^{-1} u = u_1^2 + \sum_{t=2}^{n} \frac{(u_t - \phi_t u_{t-1})^2}{1 - \phi_t^2}.$$

For a binary coupling pattern, each step from visit $t-1$ to visit t costs:

Step	Cost
on to on	$(1-\phi_t)/(1+\phi_t)$
on to off	$\phi_t^2/(1-\phi_t^2)$
off to on	$1/(1-\phi_t^2)$
off to off	0

Under graded coupling, the off-drug coupling decays instead of switching, which lowers the on-to-off cost. That is why the ceiling under graded coupling rises with the half-life: under compound symmetry, from 0.256 without carryover to 0.307 at 1 week and 0.369 at 2 weeks, which makes $c_{bm} = 0.3$ feasible from 1 week (Figure 4).

The non-separable form. With a common ρ, setting the same-visit cross-component correlation to c_1 and the different-visit correlation to $c_\times A_{ts}$ gives

$$M = K_a \otimes A + K_b \otimes I_n, \qquad K_a = (1-c_\times)I_3 + c_\times J_3, \qquad K_b = (c_1 - c_\times)(J_3 - I_3).$$

K_b has zero diagonal and so is indefinite. The eigenvalues of M are $\lambda_j(A)(1+2c_\times)+2(c_1-c_\times)$ and $\lambda_j(A)(1-c_\times)-(c_1-c_\times)$. Hence M is positive definite if and only if $\lambda_{\min}(A) > (c_1-c_\times)/(1-c_\times)$.

RH's structure. RH's compound-symmetry form is $M = K_o \otimes I_n + K_p \otimes J_n$, with a person-level factor $K_p = \rho I_3 + c_\times(J_3 - I_3)$ and an occasion-level factor $K_o = (1-\rho)I_3 + (c_1 - c_\times)(J_3 - I_3)$. It is positive definite if and only if $(1-\rho) - (c_1 - c_\times) > 0$, whatever the schedule (0.20 at RH's values).

The summed response. With component standard deviations s_t and $S_t = \sum_c s_{c,t} X_{c,t}$, separability gives

$$\mathrm{Cov}(S_t, S_s) = (s_t^\top K s_s)\, A_{ts},$$

proportional to the AR(1) kernel when the standard deviations are constant. The non-separable form adds $(s_t^\top K_b s_t)\, \mathbf{1}[t=s] > 0$, a white-noise term.

These results are derived in full, and checked numerically, in the pmsimstats-ng documentation (`docs/35-separable-response-covariance.md`).

Appendix B. Code changes

B.1 Graded coupling in RH's `generateData()` (commit 58b32a9)

The published lines (`R/generateData.R`, lines 131 to 138) couple the biomarker wherever the drug-response mean is nonzero:

```
# correlation with biomarker
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p],"br",sep=".")
  if(means[which(n1==labels)]!=0){
    correlations[n1,'bm']<-modelparam$c.bm
    correlations['bm',n1]<-modelparam$c.bm
  }
}
```

The replacement couples in proportion to the remaining drug effect, with no coupling before first exposure:

```
# correlation with biomarker: full on drug, decaying with the drug
# effect off drug (graded coupling), zero before first exposure
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p],"br",sep=".")
  if(d[p]$tod>0){
    g<-1
  }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
    g<-(1/2)^(d[p]$tsd/modelparam$carryover_t1half)
  }else{
    g<-0
  }
  correlations[n1,'bm']<-modelparam$c.bm*g
  correlations['bm',n1]<-modelparam$c.bm*g
}
```

B.2 Stop instead of repairing

The published code replaces a non-positive-definite matrix with a nearby valid one when its `makePositiveDefinite` option is on, as it is in RH's driver (lines 145 to 150). The replacement stops, so that an infeasible cell is reported rather than silently changed:

```
# published:
# if(makePositiveDefinite){
#   if(!is.positive.definite(sigma)){
#     sigma<-make.positive.definite(sigma, tol=1e-3)
#   }
# }
if(!corpcor::is.positive.definite(sigma)){
  stop(sprintf('c.bm = %.3f exceeds the feasible range for this design',
               modelparam$c.bm))
}
```

B.3 Anchored carryover on the drug-response mean

The published recursion (lines 83 to 92) decays the previous visit's mean by the total time since discontinuation. At the second and later off-drug visits that mean has already been decayed, so the decay compounds. The replacement decays from the last on-drug value. It has not been run separately in this paper. Combined with the restart of the Gompertz clock at re-exposure, it produces a downward step in the mean at the first re-exposure visit when the residual exceeds the restarted curve's value.

```
brmeans<-modgompertz(d$tod,rp$max,rp$disp,rp$rate)
if(nP>1){
  last_on<-0
  for(p in 1:nP){
    if(d[p]$onDrug){
      last_on<-brmeans[p]
    }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
```

```
        brmeans[p]<-last_on*(1/2)^(d$tsd[p]/modelparam$carryover_t1half)
      }
    }
  }
```

B.4 The exposure-decayed drug indicator in the analysis

Analyses that use an exposure-decayed drug indicator must code off-drug visits before first exposure as unexposed. The time since discontinuation is zero there, and the decay function returns 1 at zero, so without a guard a placebo-first crossover participant would be coded fully exposed at every visit. In the pmsimstats-ng carryover manuscript this guard was missing. In a paired rerun of that manuscript's crossover cells (`analysis/scripts/carryover-sensitivity/26-co-dbc-fix-paired.R`, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1$ week, 500 replicates), restoring the guard raised the power of the exposure-weighted analysis from 0.512 to 0.814, the level of the binary indicator (0.810):

```
exposure_weight <- function(Db, tsd, halflife) {
  w <- 0.5^(tsd / halflife)
  ifelse(Db == 1, 1, ifelse(tsd > 0, w, 0))
}
```

B.5 The refined process in pmsimstats-ng

The package function `buildSigma()` (`R/generateData.R`) implements graded coupling, with `c.bm` on drug, `c.bm * exp(-lambda_cor * tsd)` off drug after exposure and zero before exposure. It also implements the AR(1) structures and the placebo coupling `c.bm.pb`. Two parts of the refined process are not yet in the package:

- **The repair is still there.** Its sampler still repairs a matrix silently when the Cholesky factorization fails (`R/generateData.R`, lines 390 to 392). That repair must become an error, as in B.2.
- **The remaining refinements are in the drivers only.** The separable form, anchored carryover, and the natural-course and measurement-error options are implemented in the simulation drivers of the stress test and are to be moved into the package.